

Dissertation - Living document



Forecast USA's electrical energy production for the next
month/year

Kusal Prakash - 2021c104050

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1. Revision history

SL No	Version	Change	Comments
1	1.0	Published version	Initial published version
2	1.1	Amended version	Post review with Sujatha S Mam • Added Section 6 - Literature review • Added Section 7 - Proposed enhancements • Added Section 8 - Data Sources • Added Section 9 - Acronyms
3	1.2	Details added for the mid-semester report	Amended Sections 8 - Data sources Added Section 9, 10, 11, 12, 14

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
DSECCZG628T DISSERTATION

SECOND SEMESTER 2022- 2023

Project Work Title : Forecast USA's future electrical energy production for the next month/year

Name of Supervisor : Anirban Pramanik

Name of Student : Kusal Chand Prakash

BITS ID No. of Student : 2021C104050

2. Abstract (around 250 words):

Electrical energy is the backbone of a country, more in the case of USA as it is a superpower. It is indispensable for supporting and powering USA's critical infrastructure, industry, services, and daily activities that are essential for the functioning and the development of USA.

The law of 'Energy conservation' states that 'Energy can neither be created nor destroyed, it can only be transferred or transformed from one form to another.' Energy cannot be 'stored.'

If USA produces more electrical energy than it needs, it can face challenges such as oversupply, which can lead to wasted resources, increased costs for storage and potential strain on the electrical grid.

If USA produces less electrical energy than it needs, it can experience power shortages, leading to disruptions in daily activities, economic setbacks, and potential impacts on the quality of life for its citizens. It can also make USA dependent on energy imports hence making it vulnerable to supply disruptions, price fluctuations and geopolitical factors, potentially affecting its energy security.

Hence, the challenge for USA to produce the optimal amount of energy. To achieve this USA needs to predict with a fair degree of accuracy its annual energy needs and prepare for the same (in the long and short term).

In this dissertation we are going to use USA's historical data of electrical energy production vs consumption, population, weather, per-capita-income etc. to predict its electrical energy needs for the next month or next year.

With the right data this model could be tweaked to predict the electrical energy requirement for any other country.

Key Words (in Alphabetical order): ANN, ARIMA, ARMA, LSTM, prediction, stationary data, RNN, white noise

(Signature of Supervisor)

Name: Anirban Pramanik

Date: 14Jun2023

(Signature of Student)

Name: Kusal Chand Prakash

Date: 14Jun2023

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

DSECC ZG628T DISSERTATION

Dissertation Outline

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Designation of Supervisor: Senior Principal – Architecture

Qualification and Experience: MCA, Vellore Institute of Technology, Tamil Nadu

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Title of Dissertation: Forecast USA's future electrical energy production for the next month/year

Name of First Examiner: _____

Designation of First Examiner: _____

Qualification and Experience: _____

E- mail ID of First Examiner: _____

Supervisor's rating of the Technical Quality of this Project Work Outline

EXCELLENT / GOOD / FAIR/ POOR (Please specify): Good

Supervisor's suggestions and remarks about the outline:

The outline has been laid out very well. Kusal is using different simple to complex time series forecasting techniques to converge on a stable result.
If we can set up this experiment considering the external levers and compare the univariate Vs multivariate forecast results that would be great.

(Signature of Supervisor)

Name: Anirban Pramanik

Date: 14Jun2023

(Signature of Student)

Name: Kusal Chand Prakash

Date: 14Jun2023

3. Discussion on the chosen topic

The scope of the work is to study the historic monthly production of electrical energy in USA for the past fifty years (say) and then predict the electrical energy requirements for the next year / next month.

In addition, evaluate and study if the population, temperature, real disposable per income has any effect on the electrical consumption and hence electrical production in USA.

Real Disposable Personal Income considers changes in prices over time by adjusting for inflation, allowing for a more accurate comparison of income levels across different periods. It provides insights into the purchasing power and standard of living of individuals or households.

Even though this model is used to predict the monthly/annual electricity requirements for USA, with the right data this model could be tweaked to predict the electrical energy requirement for any other country.

4. Detailed Plan of Work (for 16 weeks)

The plan of work should have tangible weekly or fortnightly milestones and deliverables, which can be measured to assess the adherence to the plan and therefore the rate of progress in the work. The plan of work can be specified in the table given below:

Serial Number of Task	Tasks or subtasks to be done (be precise and specific)	Planned duration in weeks	Specific Deliverable in terms of the project
1	Data gathering - Define and document the data requirements, identify authentic data sources, collect data, synthesize data if required, do data quality check, data validation. Ensure adherence to data privacy and ethics	2.0	Python code + Document
2	Data pre-processing - Perform data cleaning, data integration, data transformation, feature selection, feature engineering, data normalization, handling imbalanced data, splitting data, handling time series data - stationary vs seasonality time-series-data, white noise exploration etc.	2.5	Python code + updated document
3	Data exploration and visualization - Get an initial insight from the data, univariate / bivariate / multivariate analysis, time series analysis, interactive analysis, documents the findings and insight from the visuals, iterative exploration, stationary Vs seasonal data, converting non-stationary data to stationary data. Use ACF (Auto correlation function) and PACF (Partial Auto Correlation Function) to determine the order of the models.	2.5	Python code + updated document
4	Model selection and implementation - Document the problem and the prediction requirements, identify / experiment with the different time series	3.0	Python code + updated document

	mode like ARIMA (Auto Regressive Integrated Moving Average), ARMA (Auto Regressive Moving Average), LSTM (Long Short-term Memory), RNN (Recurrent Neural Network), MAM (Moving Average Model), VAR (Vector Auto Regression), FB (Facebook) Prophet. Establish evaluation metrics and evaluate performance of the models against these metrics, do model selection, model fine-tuning.		
5	Training, tuning and evaluation of the models: Split data, do initial model selection, train the model, evaluate initial performance, hyperparameter tuning, train with tuned hyper-parameters, evaluate models with tuned hyperparameters, iterative tuning, final model selection	3.0	Python code + updated document
6	Results analysis and discussion: Assess the model performance, interpret predictions, compare with baselines, document the results, and draw conclusions	2.0	Python code + updated document
7	Final dissertation writing: Final documentation and review of the dissertation	1.0	Python code + updated document

5. Any other relevant information in support of your work.

None for now.

6. Literature review

Serial number	1
Research paper	Prediction of energy demands using Neural Network with model identification. Link
Algorithm / Technique / Model / Method used	Global optimization method called the Modal trimming method, Quasi-Newton method.
Data set used	23-week days in July 1996, sampling time interval = 1 hour.
Attributes used	Electrical energy, temperature, humidity used as supplementary conditions.
Advantages	Prediction using global optimization method is more accurate than the local optimization method.
Suggestions / Future	Only 2 dependent features used.

enhancement/ Research gap	The consideration of supplemental conditions can improve the accuracy of the identification of the values of model parameters for the neural network model, but it may deteriorate the accuracy of the prediction. Hence effect of other features like population, weather elements like HDD, CDD, RDPI on energy production cannot be determined. Does not provide an annual prediction of electrical energy.
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Serial number	2
Research paper	Time series forecasting for building energy consumption using weighted Support Vector Regression with differential evolution optimization technique. Link
Algorithm / Technique / Model / Method used	Weighted-hybrid SVR model. Parameter selection for the SVR models and optimization of the weights is done by the Differential Evolutionary algorithm.
Data set used	Electrical energy data (kWH) requirement of a building in Singapore for a year.
Attributes used	Electrical energy.
Advantages	SVR models with optimization algorithm can achieve overall high forecasting accuracy due to its effectiveness in modeling complex and nonlinear data series. The proposed model was applied to two datasets for the same building. The two datasets are that of daily energy consumption for a period of one year and half hourly energy consumption for a period of ten days. It is observed that the proposed model can forecast the energy consumption for both the datasets with good accuracy, without changing the model structure. This is achieved by using the weighted approach in which the DE algorithm can assign suitable weights to the SVR models with most impact.
Suggestions / Future enhancement/ Research gap	Univariate analysis only done. There is scope of multivariate analysis and prediction.

Serial number	3
Research paper	Forecasting monthly electric energy consumption in eastern Saudi Arabia using univariate time-series analysis. Link
Algorithm / Technique / Model / Method used	Univariate Box-Jenkins time-series analysis, ARIMA.
Data set used	Five years data.
Attributes used	Electrical energy, temperature.
Advantages	ARIMA models require less data, have fewer coefficients, and are more accurate. The optimum ARIMA model forecasts monthly data for the evaluation year with an average percentage error of 3.8% compared to 8.1% and 5.6% for the best multiple-series regression

	and Abductory Induction Mechanism (AIM) models, respectively; the mean-square forecasting error is reduced with the ARIMA model by factors of 3.2 and 1.6, respectively.
Suggestions / Future enhancement/ Research gap	Only temperature used as a dependent feature.

Serial number	4
Research paper	Monthly electric energy demand forecasting with neural networks and Fourier series. Link
Algorithm / Technique / Model / Method used	Economic variables usually influence the general series trend, while weather provides a periodic behavior because of its seasonal nature. This work investigates the periodic behavior of the Spanish monthly electric demand series, obtained by rejecting the trend from the consumption series. A novel hybrid approach is proposed: the periodic behavior is forecasted with a Fourier series while the trend is predicted with a Neural Network.
Data set used	Spanish monthly electric demand from Jan1975 to Dec2002 (a total of 336 values).
Attributes used	Past data of demand time series.
Advantages	Neural networks and curve fitting are very common techniques used for time series forecasting and specifically for load forecasting. However, the combined use of neural networks and Fourier series in electric demand forecasting is a novel approach, which has been proved to provide an adequate performance: a MAPE of 1.740% has been obtained for the whole validation period, while a value of 1.326% was obtained for the best year prediction. The correct separation of trend and fluctuation and the optimization of the forecasting tools used to carry out the predictions are the key issues in the success of this technique.
Suggestions / Future enhancement/ Research gap	Univariate analysis only done. There is scope of multivariate analysis and prediction.

Serial number	5
Research paper	Forecasting energy consumption using ensemble ARIMA-ANFIS hybrid algorithm. Link
Algorithm / Technique / Model / Method used	ARIMA - ANFIS model.
Data set used	Iran electrical energy.
Attributes used	Past data of demand time series.
Advantages	Special linear and nonlinear models can present desirable results with high correctness, especially when both models have good forecasting strength. ARIMA model can forecast linear part of data, ANFIS model can forecast nonlinear part, the hybrid model could increase forecasting results more efficiently. In this case the hybrid

	pattern that uses AdaBoost method with Genfis3 ANFIS structure and back propagation training algorithm has better results and lowest MSE value.
Suggestions / Future enhancement/ Research gap	In future studies, new diversification methods as well as using other prediction model such as Support Vector Machines can be used to improve results. Also, finding the best values of AdaBoost method parameters, with a powerful method such as genetic algorithm can make remarkable results. Moreover, each of inputs can be forecasted by ARIMA model and results used as the input of hybrid models.

7. Proposed enhancements at a very high level in this dissertation

In addition to the details mentioned in the outline the following is planned to be covered in this dissertation: -

1. None of the papers above is used to determine the electrical energy requirements of an entire country. This dissertation aims to do so.
2. Most of the prediction is done using univariate analysis. Factors that affect the electricity consumption/generation like population, multiple weather elements, RDPI have not been considered in the model. However, these factors are important and we need to study the effect of these factors in the electricity consumption/generation.
3. Use ensemble techniques (bagging and boosting) to predict the various independent and dependent features.
4. Use hybrid model - Predict the weather elements, electricity consumption, population, Real Disposable Personnel Income. Use the predicted values of these features as an input to a model to predict the electricity production.

8. Data sources

Data	Source
Monthly electricity consumption and generation in USA for the past fifty years	https://www.eia.gov/totalenergy/
Monthly weather elements - average temperature, max temperature, min temperature, HDD, CDD, precipitation for USA for the past fifty years	Data is present in the form of flat files. Data extraction and data manipulation activities need to be done. Data is present state / region wise. https://www.ncei.noaa.gov/

9. Details of task accomplished so far

a. Data source locations

Sl No	Data	Source	Comments
1	Monthly electricity consumption and generation in USA for fifty years	US Energy Information Administration	Monthly energy consumption in USA for fifty years from Jan 1973 until Dec 2022 has been used. Energy units = million kilo Watt Hours (kWH)
2	Monthly USA Real Disposable Personal Income (RDPI) for fifty years, Jan1973 until Dec2022	FRED economic data	RDPI (Real Disposable Personal Income, per capita) refers to the inflation-adjusted disposable personal income per person. It represents the amount of money available to an individual or household after deducting taxes and adjusting for inflation. Per capita means that the income is divided by the total population to provide an average income per person. RDPI unit = USD
3	Monthly temperature from Jan1973 until Dec2022	Lincon weather and climate	Temperature reading for Lincon weather was used. Temperature unit = °F
4	Monthly population from Jan1973 until Dec2022	FRED economic data	Civilian noninstitutional population is defined as persons 16 years of age and older residing in the 50 states and the District of Columbia, who are not inmates of institutions (e.g., penal, and mental facilities, homes for the aged), and who are not on active duty in the Armed Forces. The series comes from the 'Current Population Survey (Household Survey)' Unit = Thousands of Persons

b. Data pre-processing

- There were no NULL values in the data

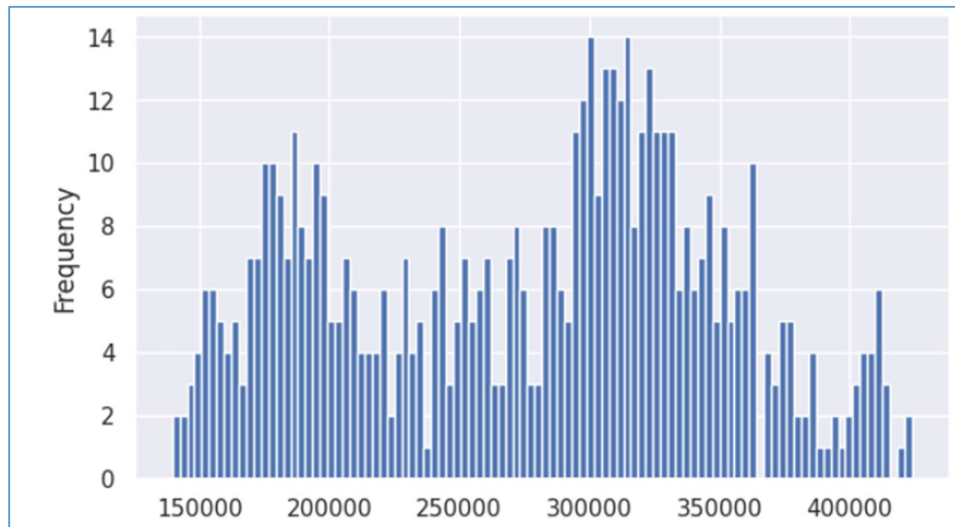
```
1 # Check for null values in all columns
2 null_values = df_gen.isnull().sum()
3
4 # Display the count of null values for each column
5 print(null_values)
```

```
month      0
coal       0
petroleum  0
n_gas      0
o_gas      0
nuclear    0
p_hydel    0
c_hydel    0
wood       0
waste      0
geothermal 0
solar      0
wind       0
tot_genm   0
dtype: int64
```

```
1 # Check for null values in all columns
2 null_values = df_con.isnull().sum()
3
4 # Display the count of null values for each column
5 print(null_values)
```

```
month      0
residential 0
commercial 0
industrial 0
transportation 0
cust_total 0
direct     0
tot_conm   0
dtype: int64
```

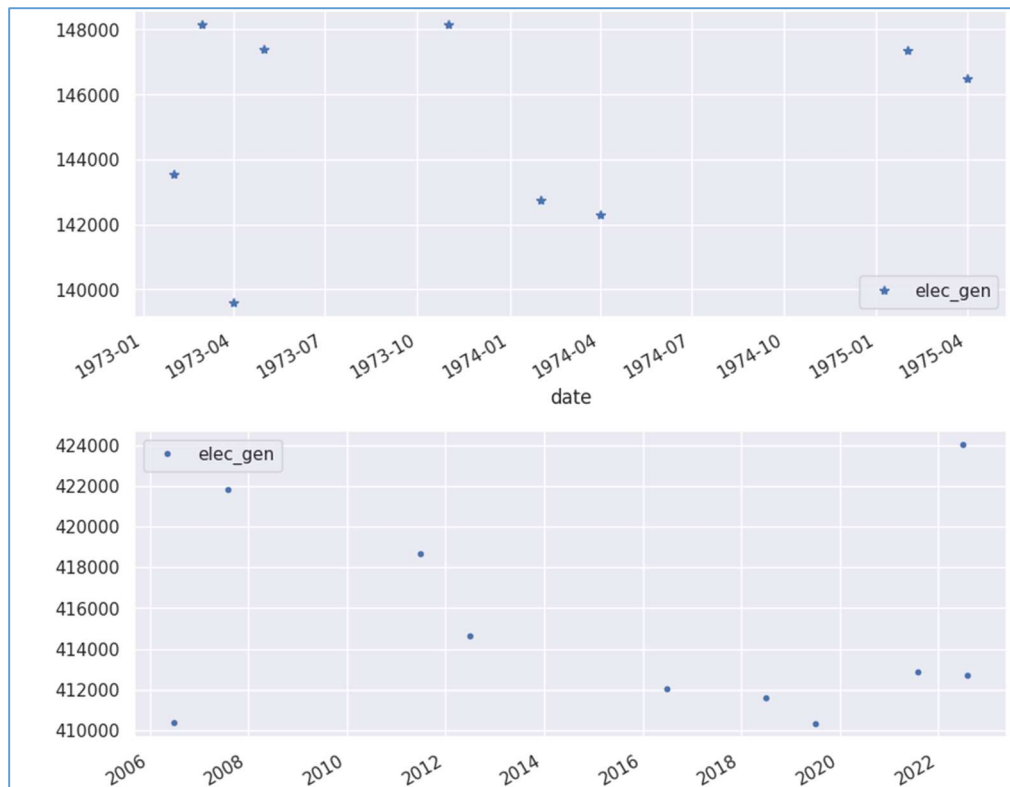
- Outlier detection



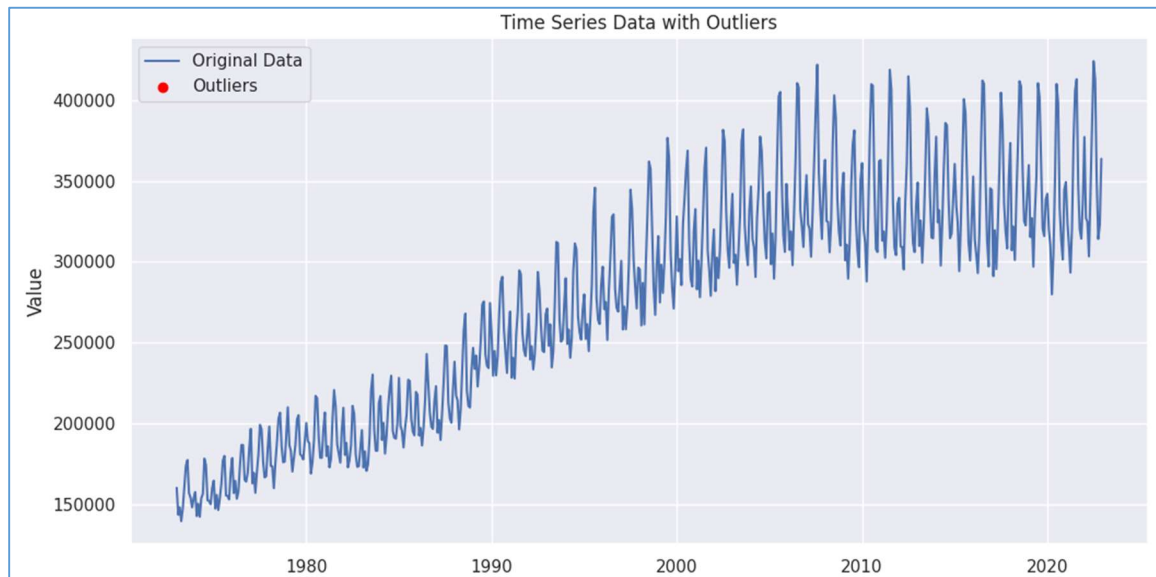
1. The values of electricity generation are min value = ~150k, max value = ~430k
2. 300k to 350k is the most generated electricity value
- 3.

```
1 df_outlier.query('elec_gen < 150000').plot(figsize = (10,4), style = '*')
2 df_outlier.query('elec_gen > 410000').plot(figsize = (10,4), style = '.')

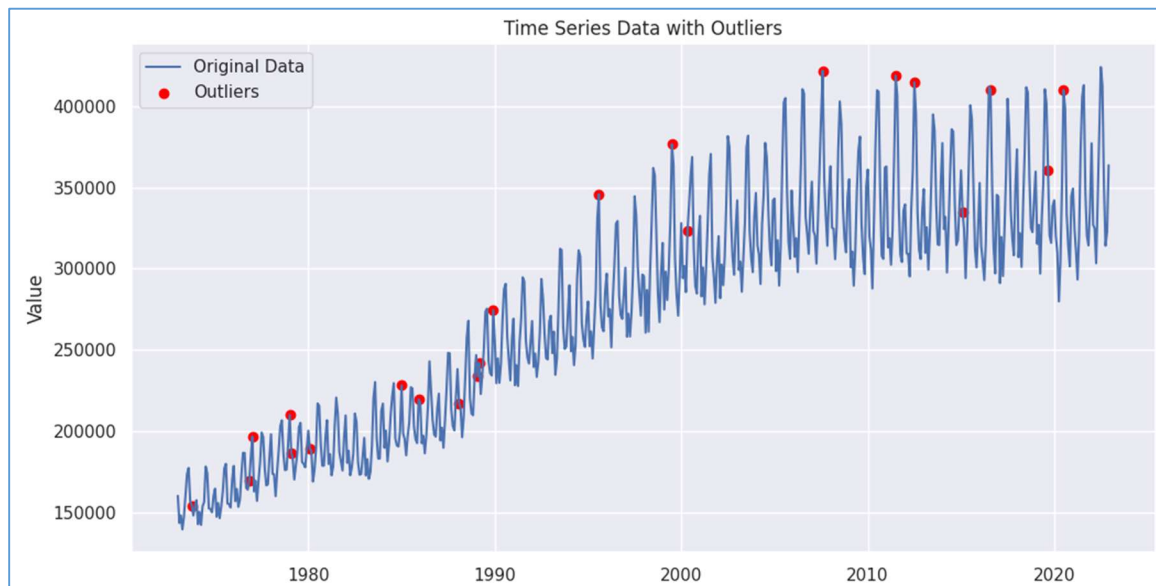
```



Here we decompose the given data and then use the residual component to check if any of the residual values exceed the threshold.

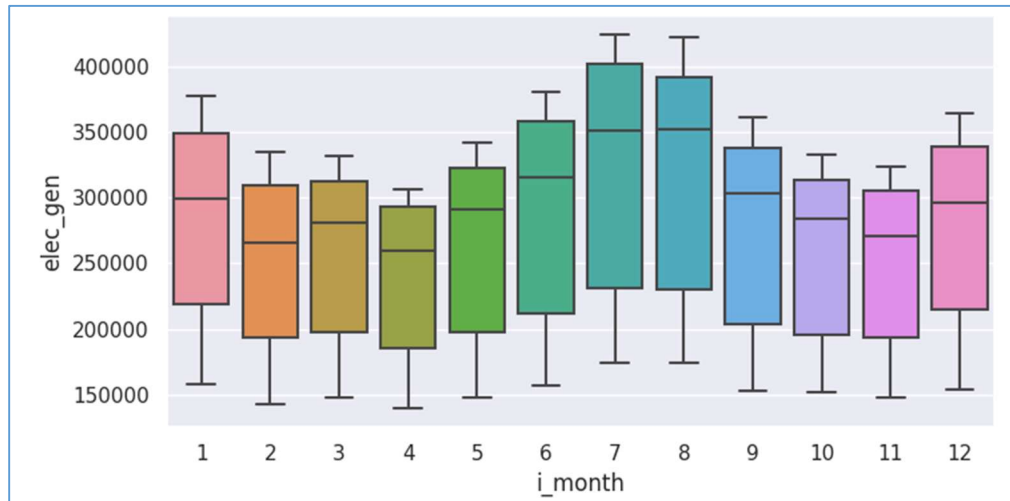


We do not see any outliers if we use the threshold = 2.5 (this is the normal threshold value) However, if we decrease the threshold value to 1.05, we do outliers.



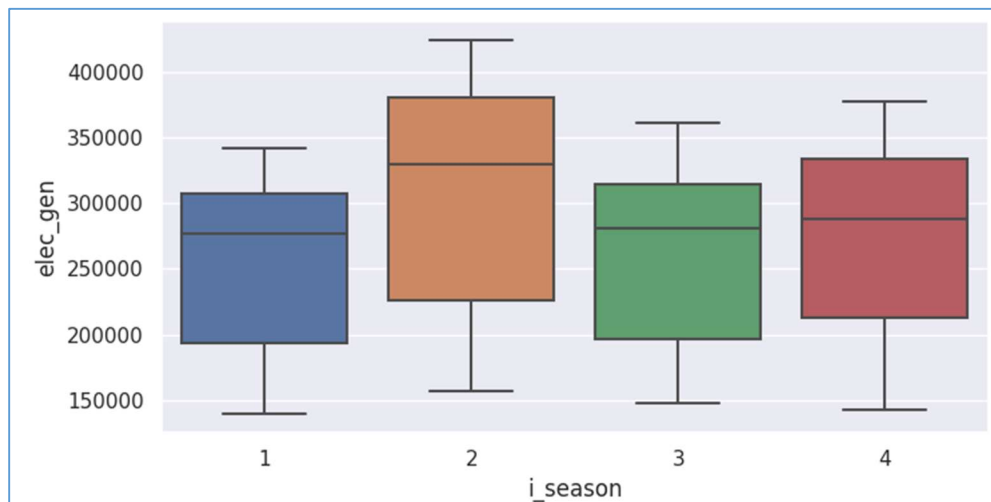
For the above graphs we can infer that there are no outlier values. The data is clean.

- Feature extraction



Inference

- Every year there are 2 peaks of electricity generation, Jan, and Jul.
- Jan - heater to heat. Jul - AC to cool.
- Lowest consumption in and around Apr, Oct.
- Hence let us create features from the given data



```

1 def month_season(month):
2     if month in [3, 4, 5]:
3         return 1 #spring
4     elif month in [6, 7, 8]:
5         return 2 #summer
6     elif month in [9, 10, 11]:
7         return 3 #fall
8     else:
9         return 4 #winter

```

Inference

- Electricity consumption is max in summer (AC to cool) and then in winter (heaters to heat)
- Hence let us create seasons and months from the given time series data

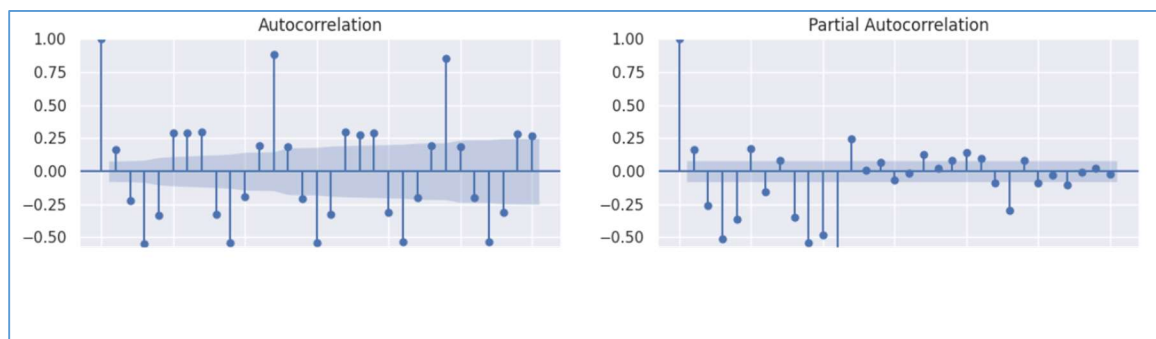
```
1 p_value(df_ts5['elec_gen'])
```

P-Value : 0.45074797868417926 . It is a non-stationary data

The given tsd has non-stationary data. Before we do ACF, PACF calculations we need to make the tsd stationary.

```
1 df_as['y1'].plot()
2 p_value(df_as['y1'].dropna()) # need to drop the NaN values else it will give error while calculating the p_value
3 print('mean = ', df_as['y1'].mean().round(2), ' and variance = ', df_as['y1'].var().round(2))
```

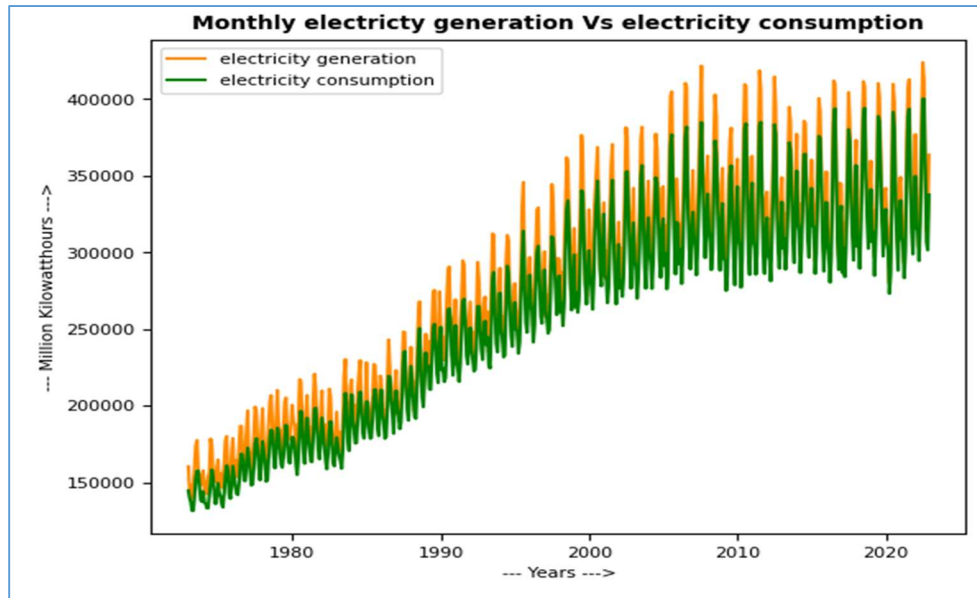
P-Value : 7.782045618059567e-13 . It is a stationary data
mean = 339.58 and variance = 667150727.96



Inference & next steps

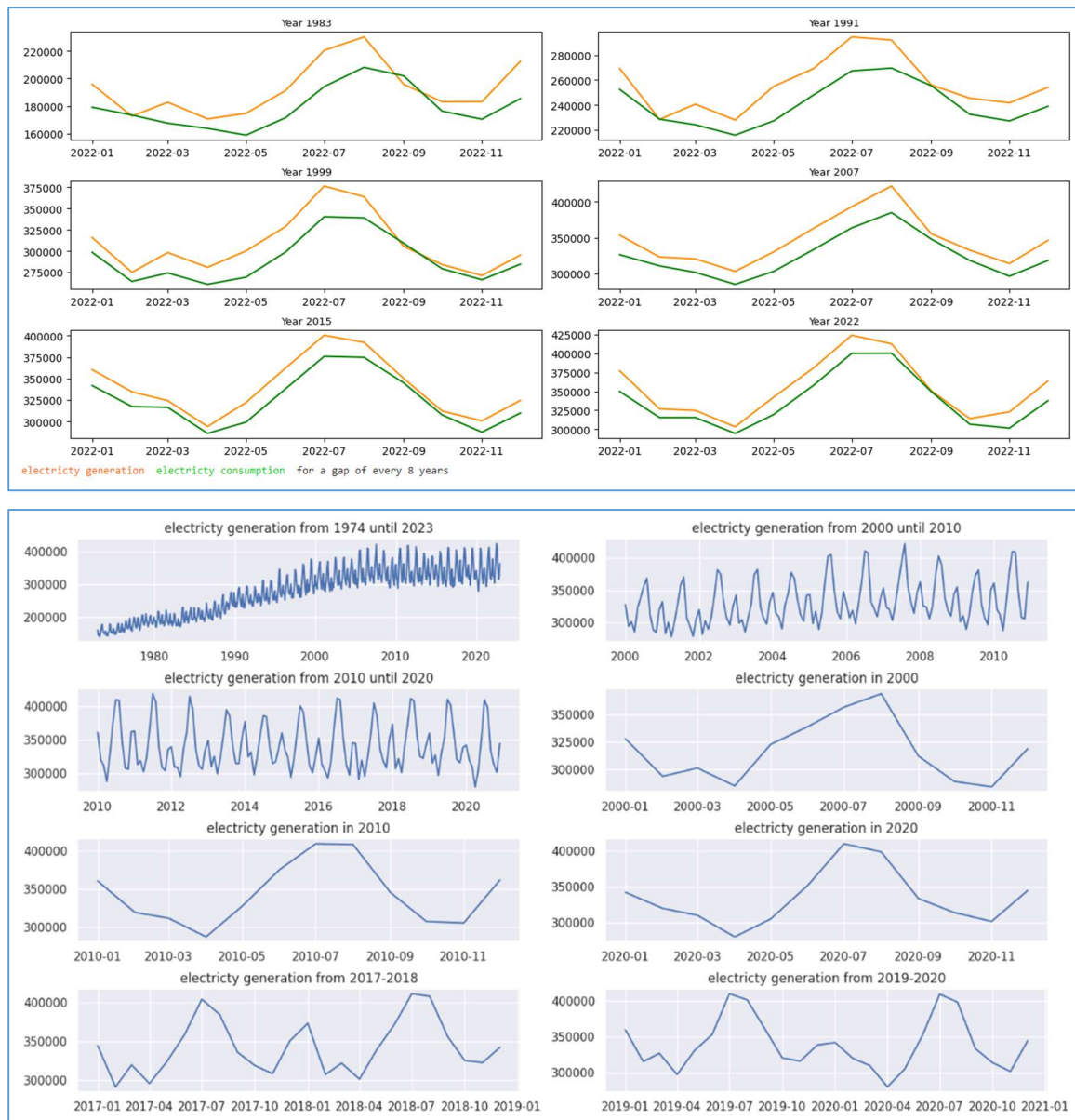
- ACF gives the direct and the indirect correlation b/w all the lags
- PACF gives the direct correlation b/w the lags
- in the above graphs x-axis is the lags, y-axis is the Pearson correlation
- from the Partial Autocorrelation graph we infer that $p = 11$
- from the Autocorrelation graph we infer that $q = 12$
- In ARIMA model for the parameters p, d, q we will use $d = 1$
- We will start from 1, 1, 1 and go to 12, 1, 12 to see the best values of rmse, mae etc.

c. EDA – Exploratory data analysis, data exploration and visualization



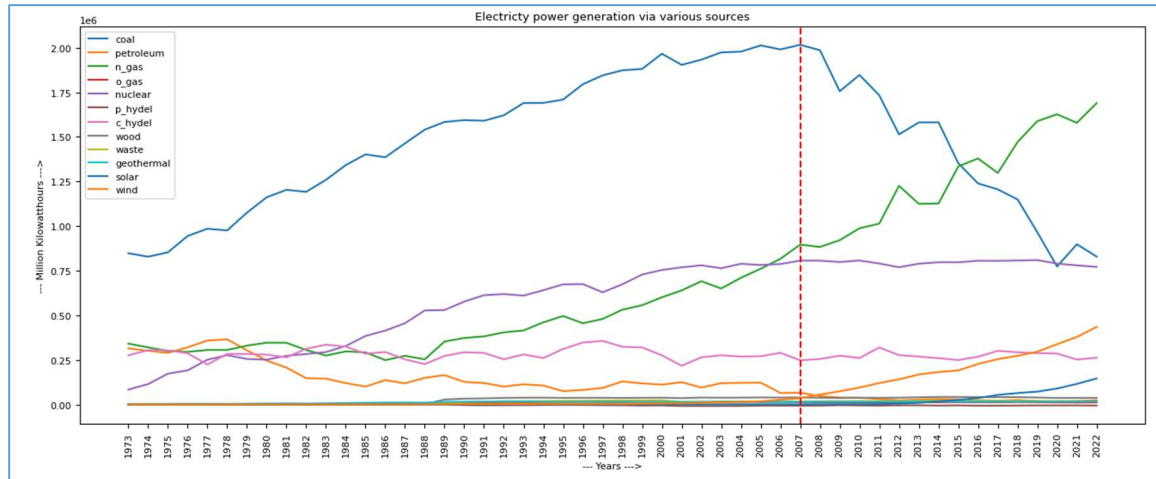
Inference

1. Over the years we see that the electricity generation is always more than the electricity consumption which is what it should normally be
2. However, during a year there are several crests and troughs which need to be studied
3. Over the years the electricity consumption has increased and so has the electricity generation



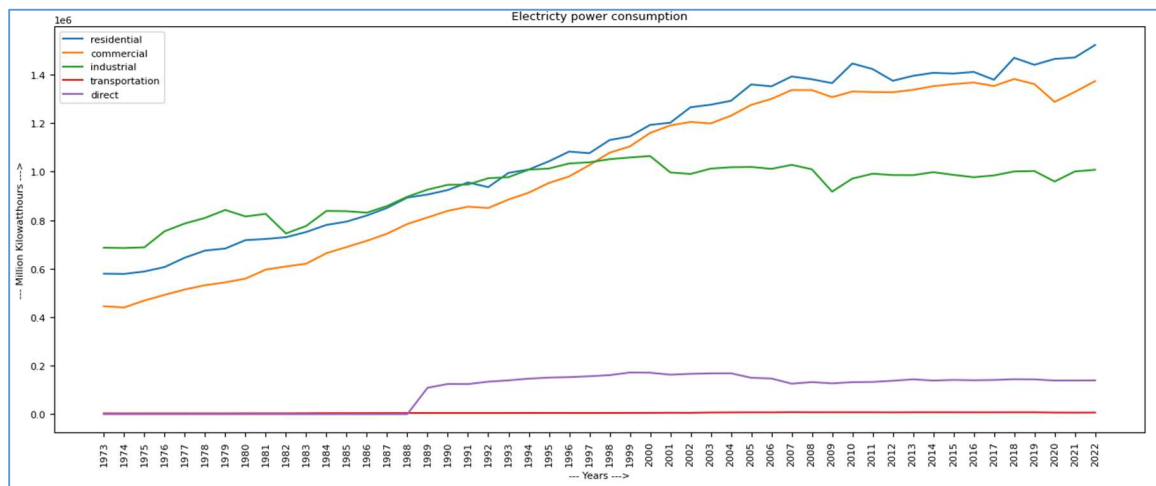
Inference:

1. We see that there has been consistent more electricity generation vs electricity consumption for the 6 years 1983/91/99, 2007/15/22
2. During Sep for most of the years the electricity generation Vs electricity consumption has been neck to neck. In 1983, 1999 and in 2022 the electricity generation was less than electricity consumption. This needs to be avoided.
3. Consistent spikes in generation and consumption during Jul and Dec months



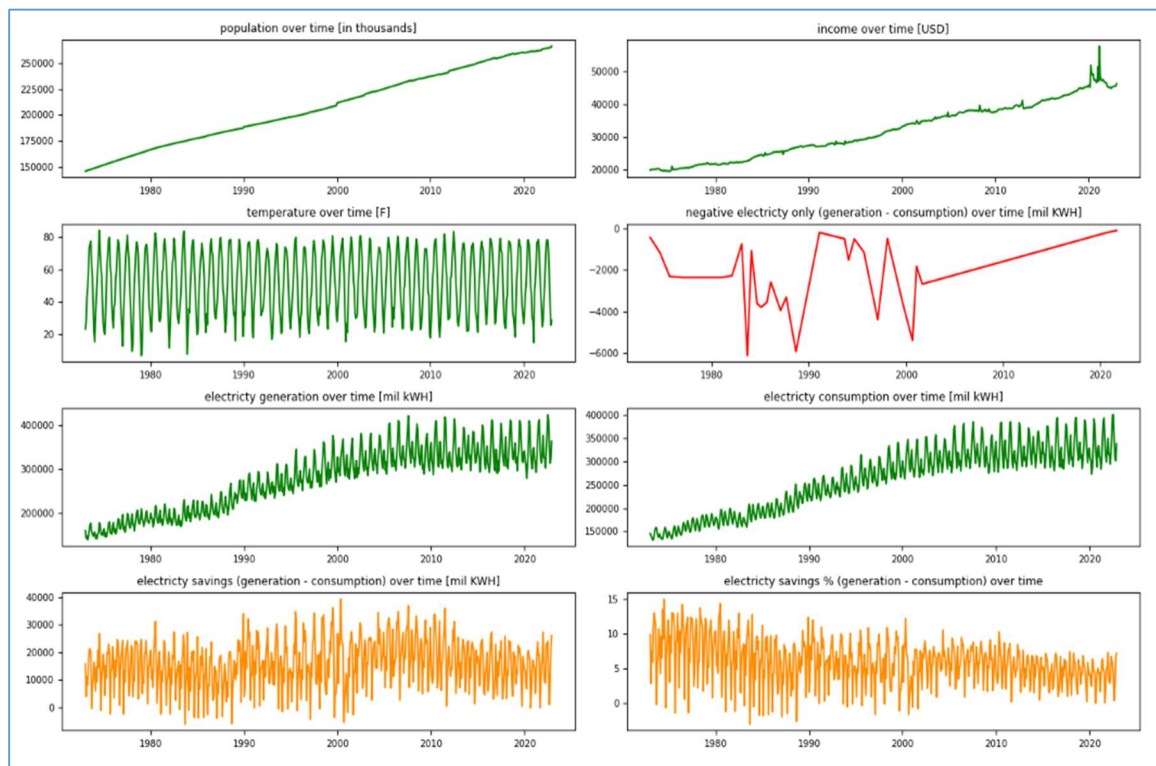
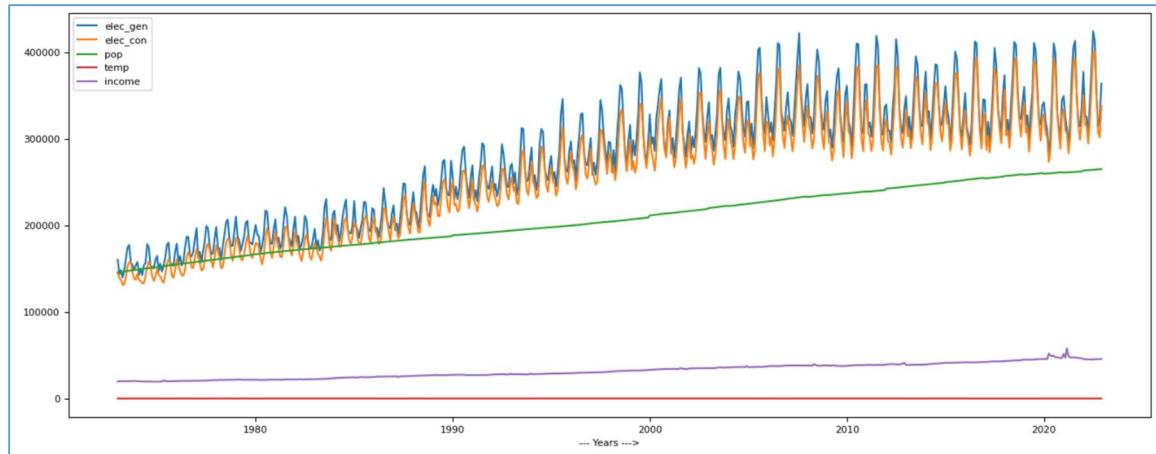
Inference:

1. Dependency on coal energy has been significantly reduced from 2007 when coal reached its peak
2. Dependency on natural gas energy has been considerably rising



Inference:

1. Residential followed by Commercial consume most of the electrical energy produced
2. Any deficiency in electricity production will cause maximum dis-comfort to the residents followed by discomfort and inconvenience to offices, retail stores, restaurants, hotels, hospitals, educational institutions, and other non-residential buildings.
3. Electricity consumption by the industries has been fairly remained constant since 1995
4. Transportation uses the least amount of electrical energy when compared to other energy consumers



Inference

- Population over time shows a consistent rising trend
- Income over time shows a consistent rising trend, with a spike post the year 2020
- Temperature over time has been constant 0-to-80-degree Fahrenheit (-17 to around 30 degrees Celsius)
- Since 1973 there has been multiple instances wherein electricity consumption has been more than electricity generation. There have been some serious spikes around the years 1984, 1988, 2002, However, after 2002 consistent effort has been made to reduce the gap to zero
- Electricity generation and electricity consumption shows a seasonal and increasing trend over time
- Electricity savings i.e., electricity generation - electricity consumption has been good over time. Max is around 15% and min is less than -5%.



Inference:

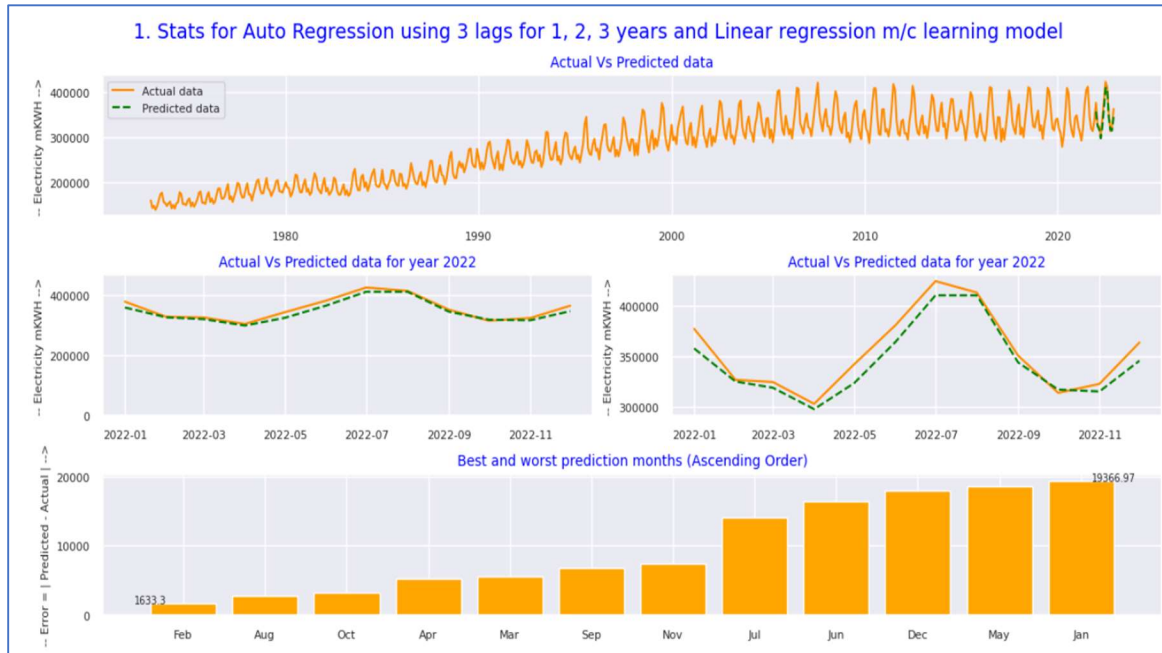
- Electricity generation has a very strong (100%) correlation with
 - electricity consumption
 - population
 - income
- Electricity generation has quite a good correlation with the temperature

Definition of the metrics used in the evaluation of the models

1. Mean Absolute Error (MAE) - measures the average absolute difference between the predicted values and the actual values. It provides a measure of the average magnitude of errors without considering the direction of the errors. **Lower MAE indicates better performance.**
2. Root Mean Squared Error (RMSE) - It is like MAE but penalizes larger errors more extremely due to the squaring operation. It provides an overall measure of the model's prediction accuracy. **Lower RMSE indicates better performance.**
3. Mean Absolute Percentage Error (MAPE) - measures the average percentage difference between the predicted values and the actual values. It provides a relative measure of the error and is useful when you want to assess the accuracy in terms of percentage. **Lower MAPE indicates better performance.**
4. R-squared error of Coefficient of determination (R^2) - measures the proportion of variance in the dependent variable (actual values) that is predictable from the independent variable (predicted values). It ranges from 0 to 1. **Higher values indicate better prediction performance.**
5. Mean Forecast Error (MFE) - measures the average difference between the predicted values and the actual values. **Positive MFE indicates over-prediction. Negative MFE indicates under-prediction. MFE close to zero indicates better performance.**

10. Strategy

a. Strategy 1: Auto regression, lag (1, 2, 3 years), m/c learning model – LR



Used 3 different types of lags

- 1, 2, 3 months
- 3, 6, 9 months
- 12, 24, 36 months

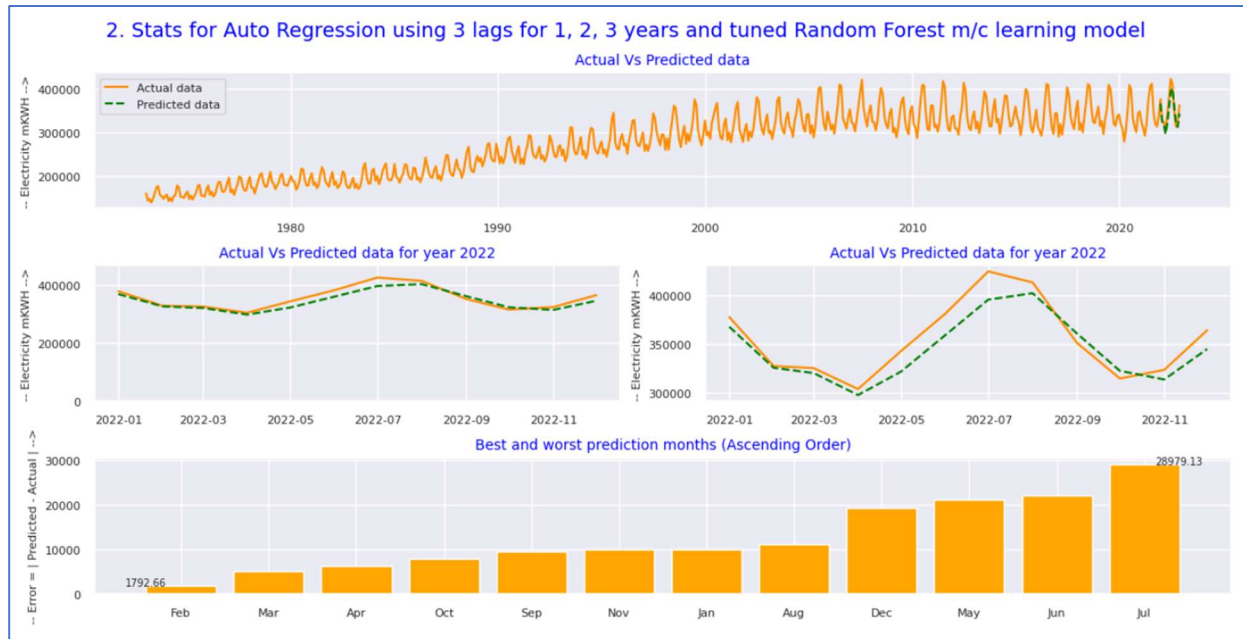
Lags with 12, 24, 36 months gave the best performance metrics for both linear regression and random forest

Sl No	Method	mae	rmse	mape	r2	mfe	Best & worst month prediction
1	Auto regression with lags (1,2, 3 years) with linear regression	9,678.100	12,762.335	2.881	0.8847	-1,520.387	Feb - 1,633 Jan - 19,367

Inference

- Negative mfe shows under fitting
- r2 b/w 0 to 1. Higher r2 shows better prediction performance
- Lower the mae, rmse, mape values better the prediction performance
- Best month prediction = Feb, worst month prediction = Jan

b. Strategy 2: Auto regression, lag (1, 2, 3 years), m/c learning model – RF



Used the different values for the below parameters to get the most optimal solution

- Number of estimators = 10, 50, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000
- Depth = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
- Max features = None, "sqrt", "auto", "log2", 0.001, 0.01, 0.5, 0.1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13
- Best month prediction = Feb, worst month prediction = Jul

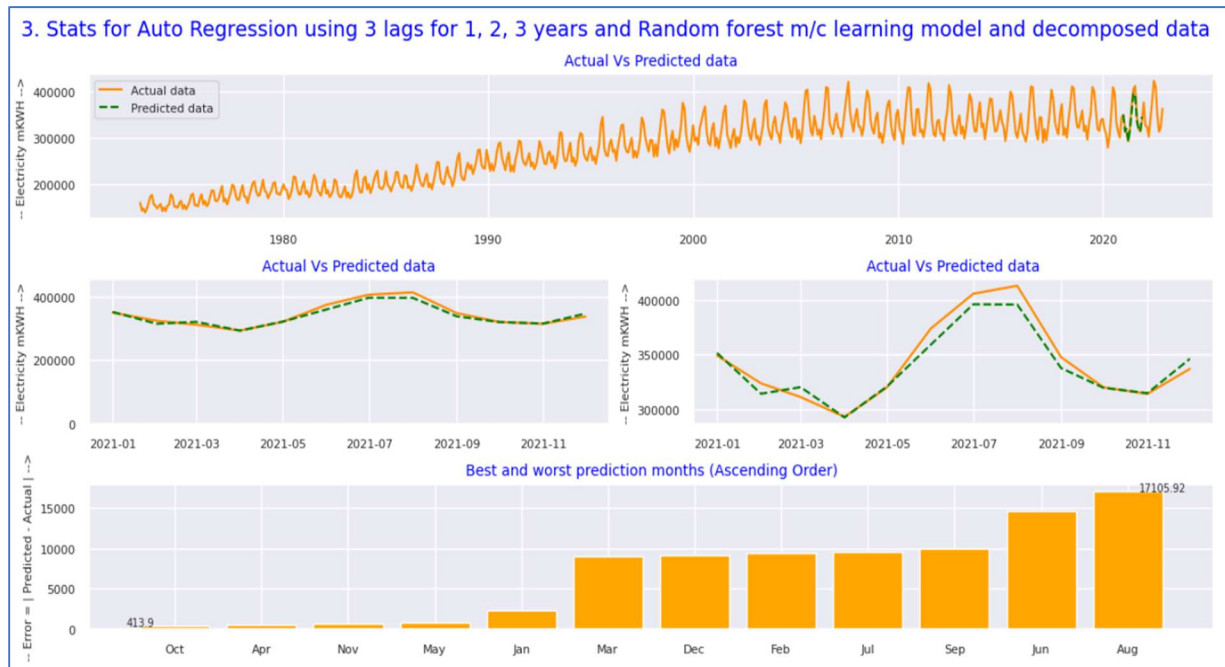
Number of estimators = 100, max_features = 0.01, depth = 12 gave the best performance metrics

Sl No	Method	mae	rmse	mape	r2	mfe	Best & worst month prediction
1	Auto regression with lags (1,2, 3 years) with linear regression	9,678.100	12,762.335	2.881	0.8847	-1,520.387	Feb - 1,633 Jan - 19,367
2	Auto regression with lags (1,2, 3 years) with random forest	38,517.641	13,567.878	11.117	0.8696	-12.474	Feb - 1,793 Jul - 28,979

Inference

- Negative mfe shows under fitting which is better than linear regression with the same lags
- r2 b/w 0 to 1. Higher r2 shows better prediction performance, in this case is slightly lesser than linear regression
- Lower the mae, rmse, mape values better the prediction performance, all of them are comparatively higher than linear regression

- c. Strategy 3: Decompose data, Auto regression, lag (1, 2, 3 years), m/c learning model – RF



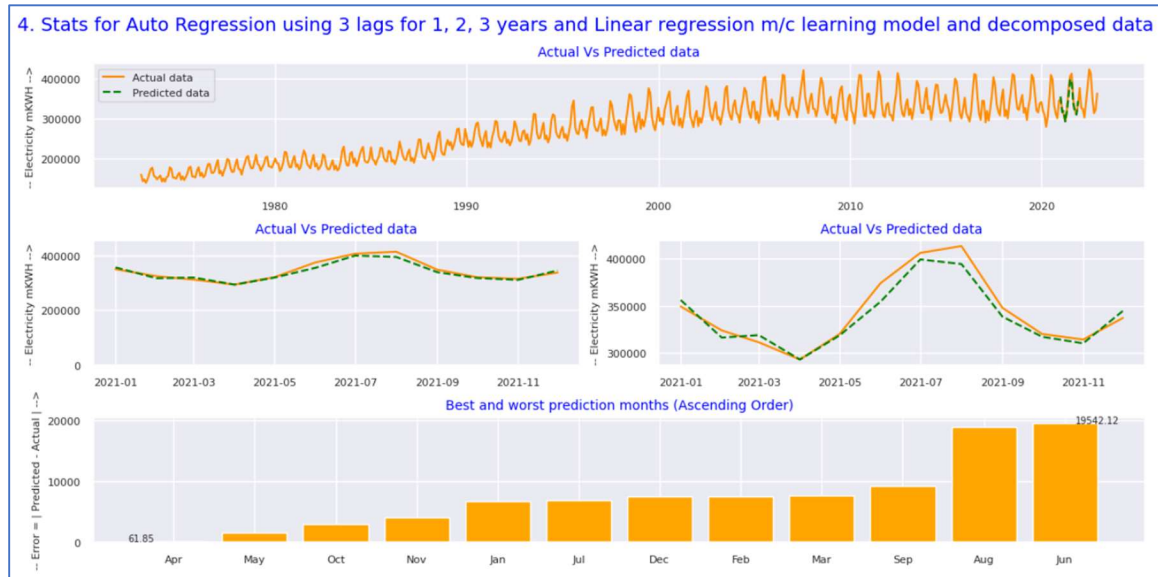
We first decomposed the data into the 3 parts viz trend, seasonal, decompose using a multiplicative model. This is since the data had a seasonal and increasing trend. Each of the 3 parts were then predicted using the RF 1, 2, 3 years lag. Post that the 3 parts were then combined to give the final predicted result.

Sl No	Method	mae	rmse	mape	r2	mfe	Best & worst month prediction
1	Auto regression with lags (1,2, 3 years) with linear regression	9,678.100	12,762.335	2.881	0.8847	-1,520.387	Feb - 1,633 Jan - 19,367
2	Auto regression with lags (1,2, 3 years) with random forest	38,517.641	13,567.878	11.117	0.8696	-12.474	Feb - 1,793 Jul - 28,979
3	Auto regression with lags (1,2, 3 years) with random forest & decomposed data	7,753.497	9,705.872	2.213	0.9282	3,488.751	Oct - 414 Aug - 17,106

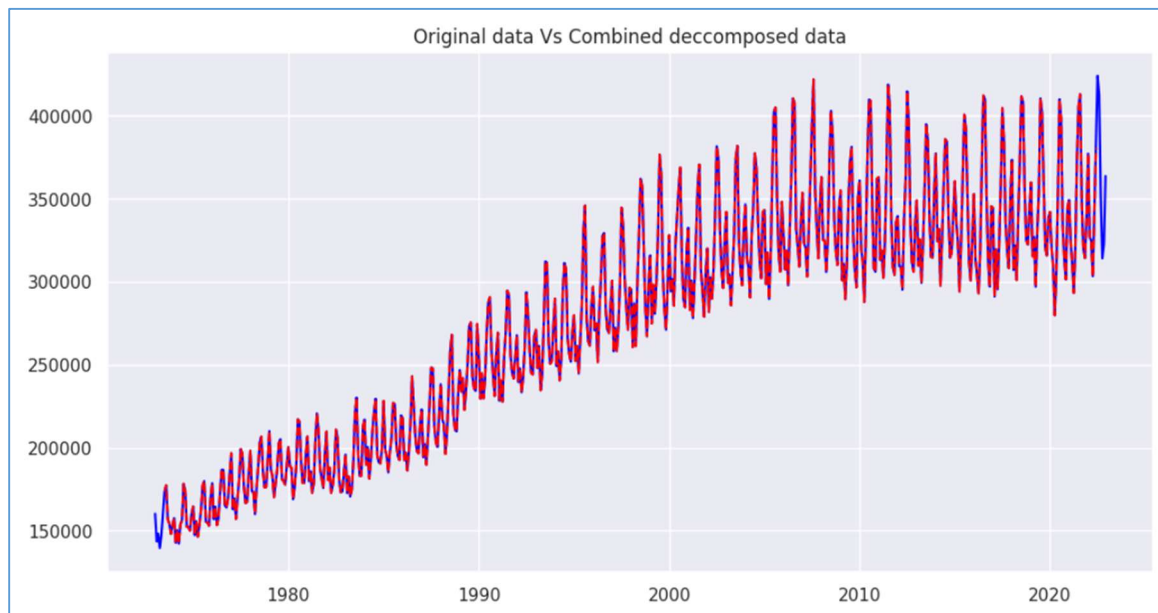
Inference

- Positive mfe shows over fitting
- r2 b/w 0 to 1. Higher r2 shows better prediction performance which is better than strategy 1 and 2
- Lower the mae, rmse, mape values better the prediction performance which is far better than the other two strategies
- Best month prediction = Oct, worst month prediction = Aug. The values have come down significantly

- d. Strategy 4: Decompose data, Auto regression, lag (1, 2, 3 years), m/c learning model – LR



We first decomposed the data into the 3 components viz trend, seasonal, decompose using a multiplicative model. This is since the data had a seasonal and increasing trend. Each of the 3 parts were then predicted using the Linear regression 1, 2, 3 years lag. Post that the 3 parts were then combined to give the final predicted result.



The decomposed values of the original data when multiplied together gives back the original value.

Inference: The predicted values of the decomposed data need to be multiplied together (not added) to arrive at the prediction.

Sl No	Method	mae	rmse	mape	r2	mfe	Best & worst month prediction
1	Auto regression with lags (1,2, 3 years) with linear regression	9,678.100	12,762.335	2.881	0.8847	-1,520.387	Feb - 1,633 Jan - 19,367
2	Auto regression with lags (1,2, 3 years) with random forest	38,517.641	13,567.878	11.117	0.8696	-12.474	Feb - 1,793 Jul - 28,979
3	Auto regression with lags (1,2, 3 years) with random forest & decomposed data	7,753.497	9,705.872	2.213	0.9282	3,488.751	Oct - 414 Aug - 17,106
4	Auto regression with lags (1,2, 3 years) with linear regression & decomposed data	9,349.552	11,946.477	2.779	0.8912	-1,301.870	Apr - 62 Jun - 19,542

Inference

- Negative mfe shows under fitting
- r2 b/w 0 to 1. Higher r2 shows better prediction performance which is second best.
- Lower the mae, rmse, mape values better the prediction performance which is again second best.
- Best month prediction = Apr, worst month prediction = Jun. The values for the best prediction have come down significantly to double digits

e. Strategy 5: ARIMA model for prediction

ARIMA (Auto Regressive Integrated Moving Average) has 3 parameters.

p - Order of Autoregression (AR): The autoregressive order is the number of past observations (lags) that are used as predictors for the current observation. It captures the r/s b/w the current observation and its previous values. Higher value of p indicates a stronger dependency on past observations.

d - Order of Differencing: The differencing order is the number of times the tsd is differenced to make it stationary. Differencing removes the trend from the tsd, making it more suitable for ARMA modeling. d = 0 if the tsd is already stationary.

q - Order of Moving Average (MA): The moving average order is the number of lagged forecast errors that are used as predictors for the current observation. It captures the relationship between the current observation and past forecast errors. Higher value of q indicates a stronger dependency on past forecast errors.

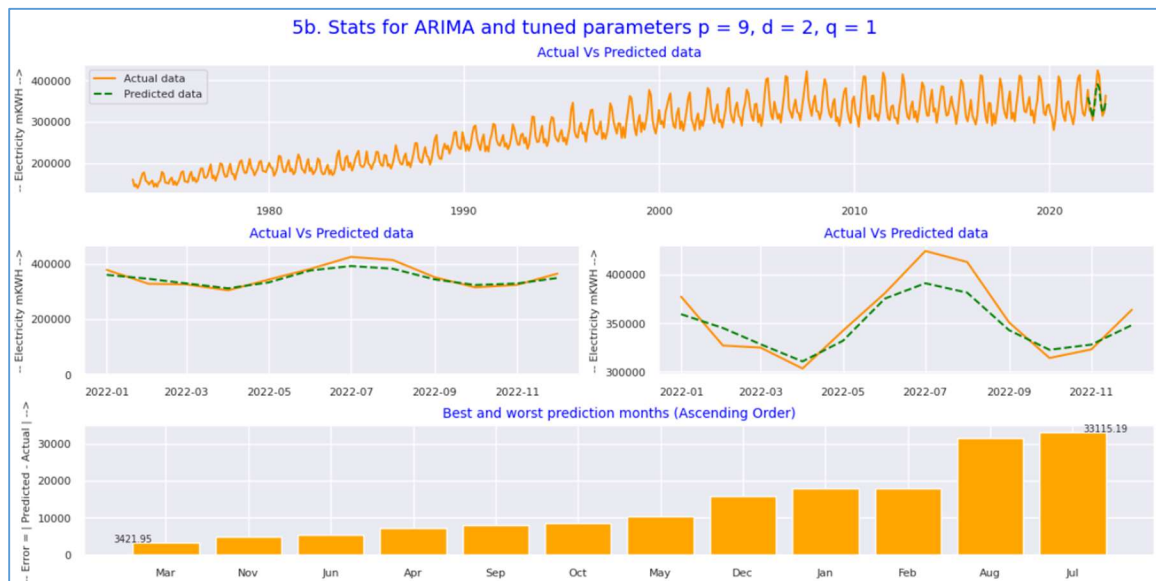
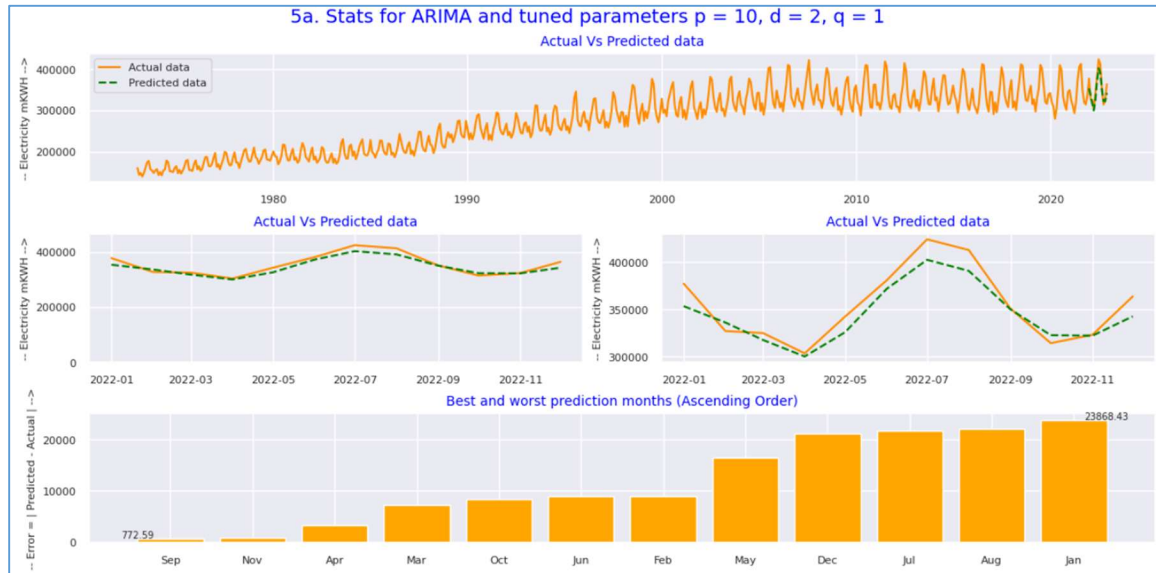
We will do fine tuning of these 3 parameters p, d, q

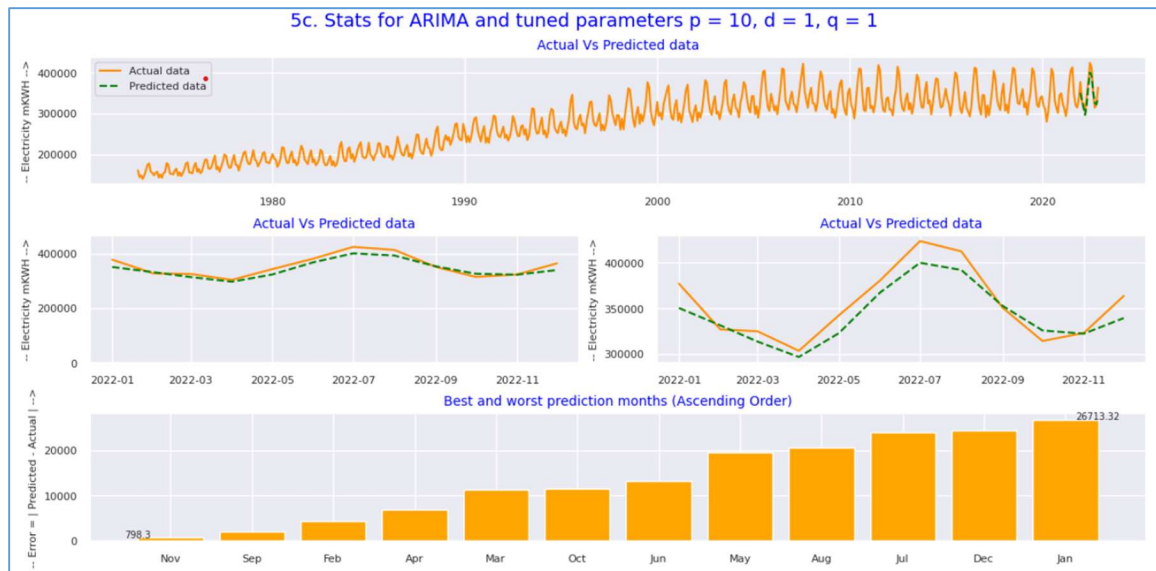
- Step 1: take d = 1 to 12 and fix the ones which give the best scores
- Step 2: for the best d; take p = 1 to 12 and fix the ones which give the best scores
- Step 3: for the best d, p; take q = 1 to 12 and fix the ones which give the best scores
- Step 4: now we have got the best values of p, d, q

Note: in each of the steps 1 to 3 when we do the prediction if we convergence coming as False then we skip the prediction step for those parameters and go to the next step. Convergence being false signifies that the model parameters used to iteratively optimize the solution is not able to trend to a solution.

After using the above permutation and combinations we get the best performance scores with the following combinations.

$(p, d, q) = (10, 2, 1), (9, 2, 1), (10, 1, 1)$





Sl No	Method	mae	rmse	mape	r2	mfe	Best & worst month prediction
1	Auto regression with lags (1,2, 3 years) with linear regression	9,678.100	12,762.335	2.881	0.8847	-1,520.387	Feb - 1,633 Jan - 19,367
2	Auto regression with lags (1,2, 3 years) with random forest	38,517.641	13,567.878	11.117	0.8696	-12.474	Feb - 1,793 Jul - 28,979
3	Auto regression with lags (1,2, 3 years) with random forest & decomposed data	7,753.497	9,705.872	2.213	0.9282	3,488.751	Oct - 414 Aug - 17,106
4	Auto regression with lags (1,2, 3 years) with linear regression & decomposed data	9,349.552	11,946.477	2.779	0.8912	-1,301.870	Apr - 62 Jun - 19,542
5a	ARIMA model (10, 2, 1)	12,006.043	14,567.790	3.258	0.8461	9,095.611	Sep - 773 Jan - 23,868
5b	ARIMA model (09, 2, 1)	13,690.137	16,696.676	3.706	0.7978	6,677.211	Mar - 3,422 Jul - 33,115
5c	ARIMA model (10, 1, 1)	13,762.636	16,293.134	3.763	0.8074	10,759.065	Nov - 798 Jan - 26,713

Inference

- Positive mfe shows over fitting for all the 3 tuned ARIMA models
- r^2 b/w 0 to 1. Higher r^2 shows better prediction performance, which is best for (10, 2, 1). However, this is the least good against all the other four models
- Lower the mae, rmse, mape values better the prediction performance - mae, mape for (10, 2, 1) ranks fourth against all the 5 models, rmse is far behind all the four models
- Best month prediction = Sep for (10, 2, 1) is third in rank (against all the five models), worst month prediction = Jan is fifth in rank (against all the five models).

When the ARIMA model was applied on the decomposed data then the results were inconclusive (convergence was coming as False). Hence, the stats are ignored.

f. Strategy 6: SARIMAX model for prediction

SARIMAX (Seasonal Auto Regressive Integrated Moving Average with eXogeneous variables) has 7 parameters. (p, d, q, P, D, Q, S)

P - Seasonal Auto Regressive order (AR): It is the number of past observations (lags) for the seasonal component of the tsd that are used as predictors for the current observation. It indicates how many lagged values of the seasonal series need to be included in the model. For example, P = 2 implies use last two seasonal data for current prediction.

D - Seasonal order of differencing: It is the number of seasonal differences required to make the tsd stationary (remove seasonality from the data).

Q - Seasonal Moving Average order (MA): It is the number of MA terms for the seasonal component of the tsd that are used as predictors for the current observation. It indicates how many lagged forecast errors of the seasonal series need to be included in the model.

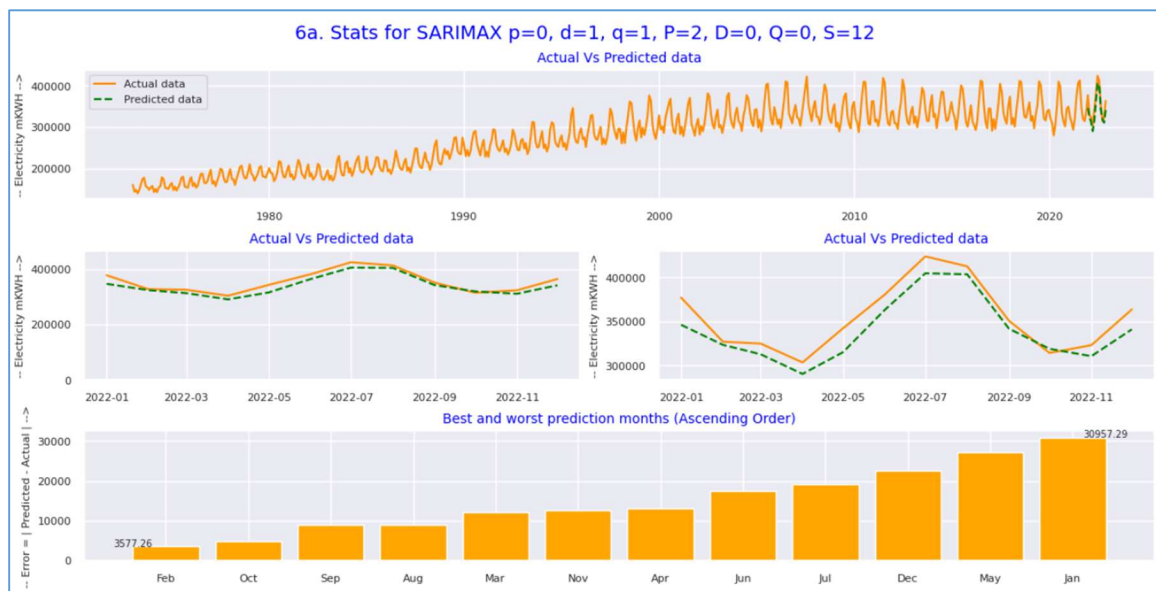
For example, Q = 1 implies use the last forecast error of the seasonal series for current prediction.

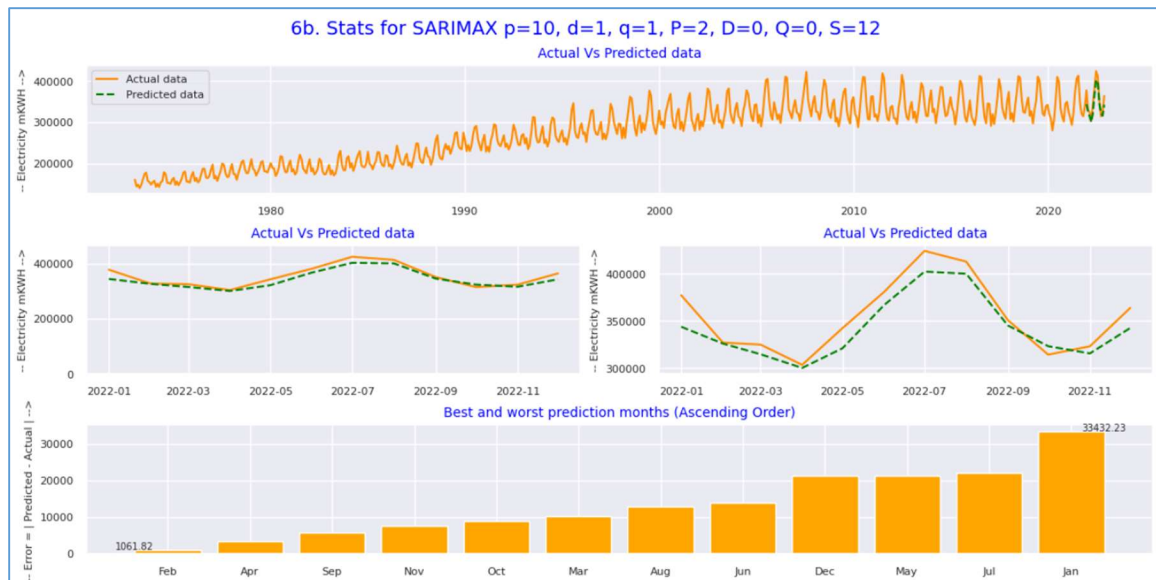
S - Seasonal Periodicity: It is the seasonal periodicity of the tsd or the number of time periods in a seasonal cycle.

For example, S = 12 implies the tsd has a monthly data and it shows seasonality over a year S=12.

More exploration is required here.

- Explore SARIMA with grid search
- Explore SARIMAX with grid search and eXogeneous variables (additional features like season, month, population, temperature, Real Disposable Personnel Income (RDPI) etc.)
- SARIMA and SARIMAX on decomposed data





Sl No	Method	mae	rmse	mape	r2	mfe	Best & worst month prediction
1	Auto regression with lags (1,2, 3 years) with linear regression	9,678.100	12,762.335	2.881	0.8847	-1,520.387	Feb - 1,633 Jan - 19,367
2	Auto regression with lags (1,2, 3 years) with random forest	38,517.641	13,567.878	11.117	0.8696	-12.474	Feb - 1,793 Jul - 28,979
3	Auto regression with lags (1,2, 3 years) with random forest & decomposed data	7,753.497	9,705.872	2.213	0.9282	3,488.751	Oct - 414 Aug - 17,106
4	Auto regression with lags (1,2, 3 years) with linear regression & decomposed data	9,349.552	11,946.477	2.779	0.8912	-1,301.870	Apr - 62 Jun - 19,542
5a	ARIMA model (10, 2, 1)	12,006.043	14,567.790	3.258	0.8461	9,095.611	Sep - 773 Jan - 23,868
5b	ARIMA model (09, 2, 1)	13,690.137	16,696.676	3.706	0.7978	6,677.211	Mar - 3,422 Jul - 33,115
5c	ARIMA model (10, 1, 1)	13,762.636	16,293.134	3.763	0.8074	10,759.065	Nov - 798 Jan - 26,713
6a	SARIMA model (0, 1, 1, 2, 0, 0, 12)	41,842.330	17,224.114	11.515	0.7848	14,346.519	Feb = 3,577 Jan = 30,957
6b	SARIMA model (1, 1, 1, 2, 0, 0, 12)	39,626.862	16,208.431	10.904	0.8094	11,968.472	Feb = 1,062 Jan = 33,432

Inference

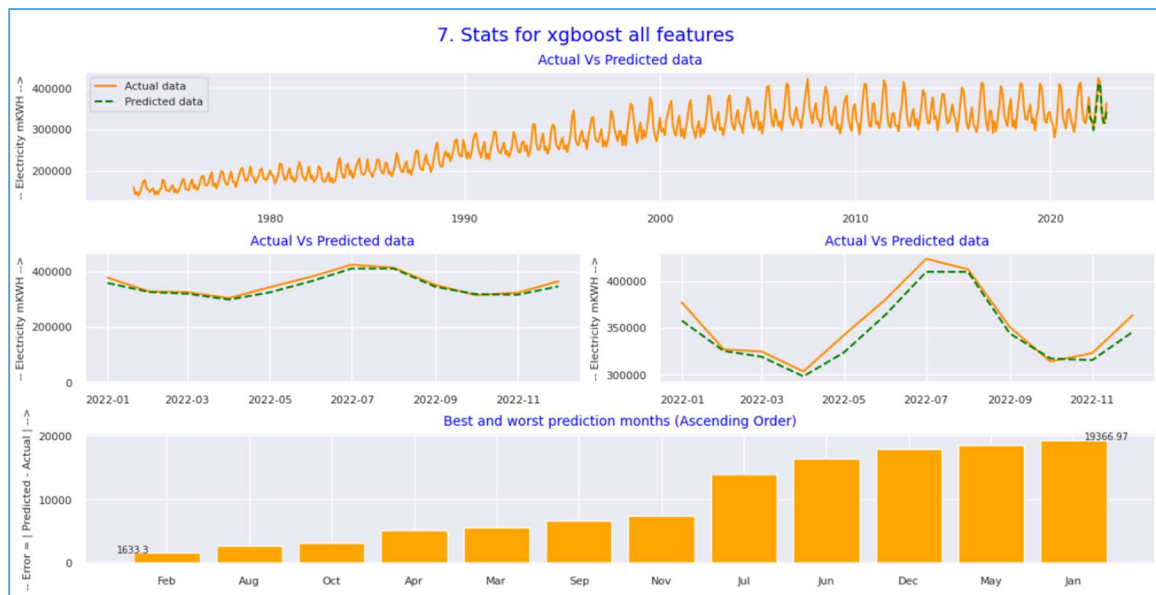
- Positive mfe shows over fitting for the 2 SARIMA models
- r2 b/w 0 to 1. Higher r2 shows better prediction performance, which is not the best when compared across all the models
- Lower the mae, rmse, mape values better the prediction performance - mae, rmse, mape is worst when compared to all the other 5 methods
- Best and worst month prediction is same for the two SARIMA models. Best - Feb and worst is - Jan.

g. Strategy 7: XGBoost (eXtreme Gradient Boost) all features

Parameters and the values used

- I. base_score = 0.5
- II. booster = 'gbtree'
- III. n_estimators = 1000
- IV. early_stopping_rounds = 50,
- V. objective = 'reg:linear'
- VI. max_dept = 3
- VII. learning_rate = 0.01

Features used - Income, population, temperature, month, season



Sl No	Method	mae	rmse	mape	r2	mfe	Best & worst month prediction
1	Auto regression with lags (1,2, 3 years) with linear regression	9,678.100	12,762.335	2.881	0.8847	-1,520.387	Feb - 1,633 Jan - 19,367
2	Auto regression with lags (1,2, 3 years) with random forest	38,517.641	13,567.878	11.117	0.8696	-12.474	Feb - 1,793 Jul - 28,979
3	Auto regression with lags (1,2, 3 years) with random forest & decomposed data	7,753.497	9,705.872	2.213	0.9282	3,488.751	Oct - 414 Aug - 17,106
4	Auto regression with lags (1,2, 3 years) with linear regression & decomposed data	9,349.552	11,946.477	2.779	0.8912	-1,301.870	Apr - 62 Jun - 19,542
5a	ARIMA model (10, 2, 1)	12,006.043	14,567.790	3.258	0.8461	9,095.611	Sep - 773 Jan - 23,868
5b	ARIMA model (09, 2, 1)	13,690.137	16,696.676	3.706	0.7978	6,677.211	Mar - 3,422 Jul - 33,115
5c	ARIMA model (10, 1, 1)	13,762.636	16,293.134	3.763	0.8074	10,759.065	Nov - 798 Jan - 26,713
6a	SARIMA model (0, 1, 1, 2, 0, 0, 12)	41,842.330	17,224.114	11.515	0.7848	14,346.519	Feb = 3,577 Jan = 30,957
6b	SARIMA model (1, 1, 1, 2, 0, 0, 12)	39,626.862	16,208.431	10.904	0.8094	11,968.472	Feb = 1,062 Jan = 33,432
7	XGBoost with all features	40,745.896	11,848.196	11.278	0.8982	9,362.162	Feb = 1,633 Jan = 19,367

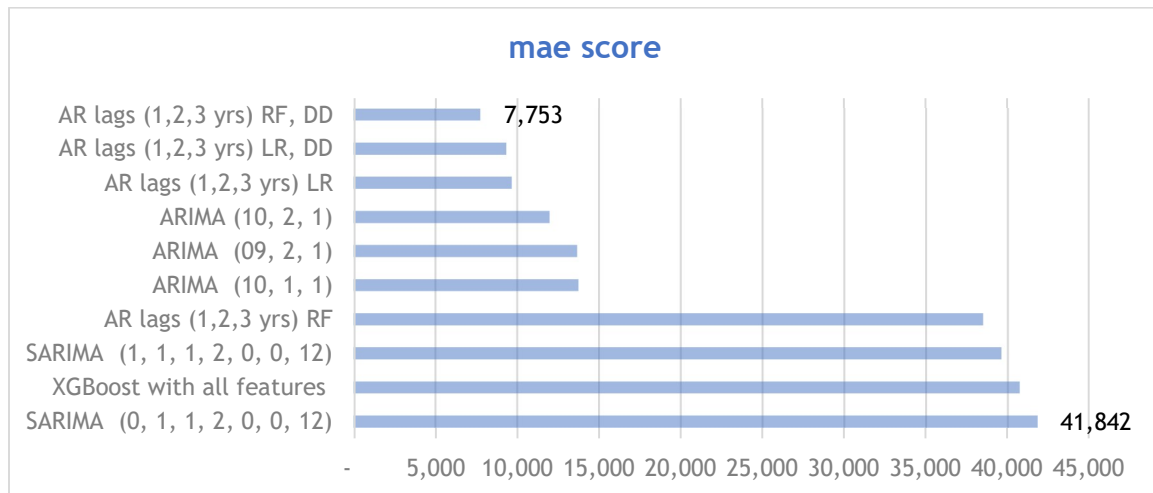
Inference

- Positive mfe shows over fitting for the XGBoost model
- r2 b/w 0 to 1. Higher r2 shows better prediction performance, which is second best when compared across all the six models
- Lower the mae, rmse, mape values better the prediction performance - mae is highest (worst across all models), rmse is third best, mape is worst when compared to all the other 5 methods
- Best and worst month prediction is same as that of the two SARIMA models. Best - Feb and worst is - Jan. However, worst month score is better than the SARIMA models

11. Analysis of the various models, performance stats

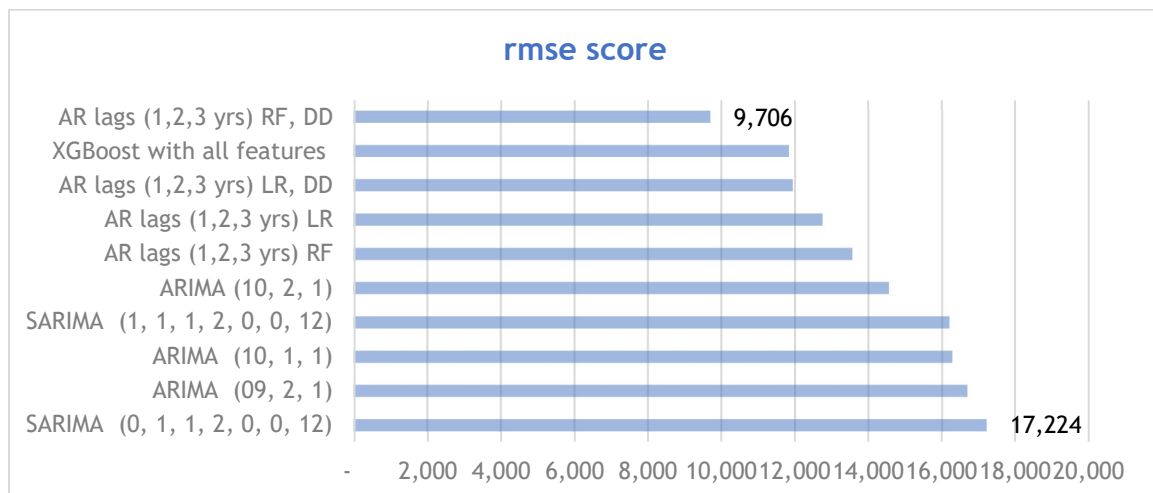
Mean Absolute Error (MAE) - It measures the average absolute difference between the predicted values and the actual values. It provides a measure of the average magnitude of errors without considering the direction of the errors. **Lower MAE indicates better performance.**

$$MAE = \frac{\sum_{i=1}^n |y_i - x_i|}{n}$$



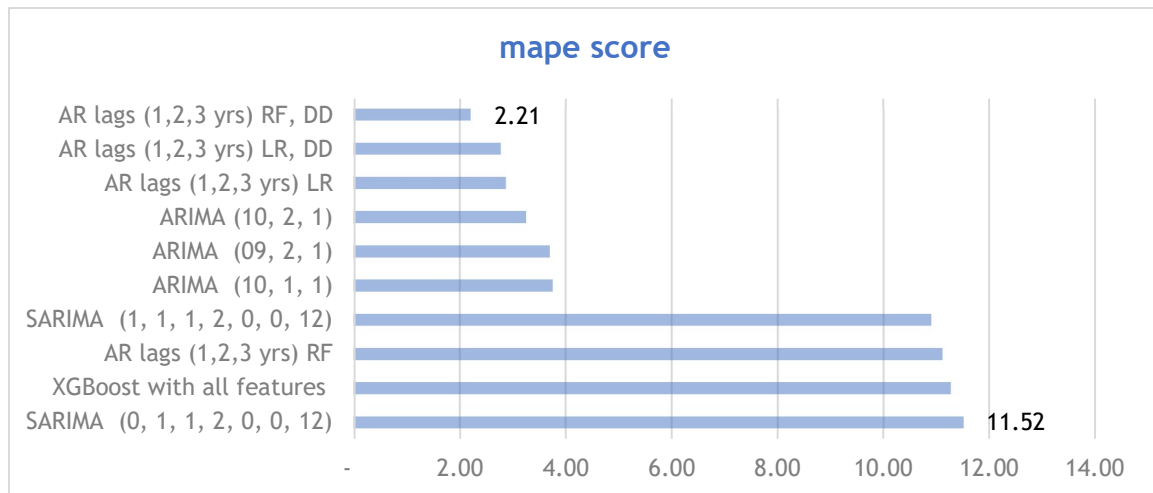
Root Mean Squared Error (RMSE) - It is like MAE but penalizes larger errors more extremely due to the squaring operation. It provides an overall measure of the model's prediction accuracy. **Lower RMSE indicates better performance.**

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (Predicted_i - Actual_i)^2}{N}}$$



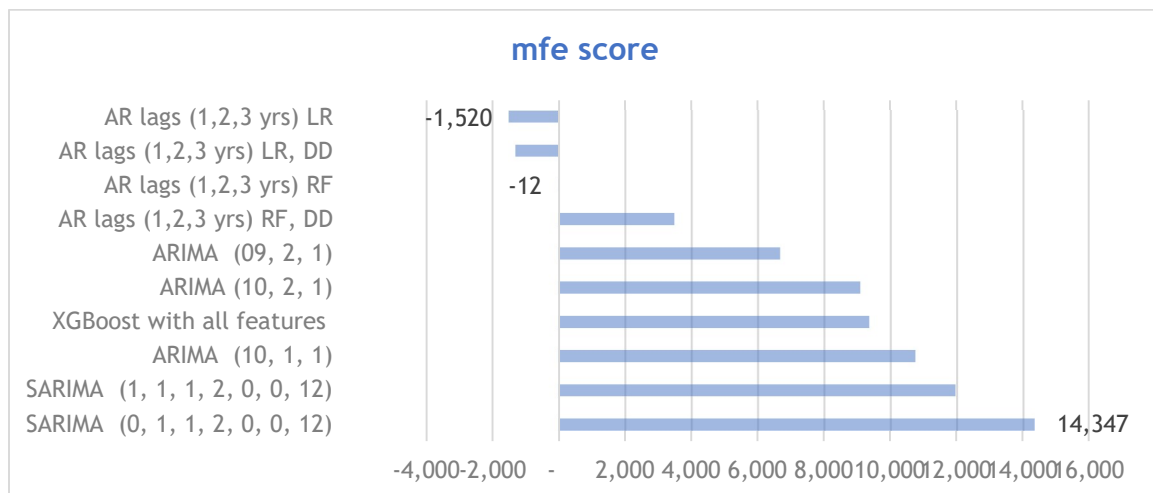
Mean Absolute Percentage Error (MAPE) - measures the average percentage difference between the predicted values and the actual values. It provides a relative measure of the error and is useful when you want to assess the accuracy in terms of percentage. **Lower MAPE indicates better performance.**

$$MAPE = \frac{1}{n} \times \sum \left| \frac{\text{actual value} - \text{forecast value}}{\text{actual value}} \right|$$



Mean Forecast Error (MFE) - measures the average difference between the predicted values and the actual values.

- **Positive MFE indicates over-prediction.**
- **Negative MFE indicates under-prediction.**
- **MFE close to zero indicates better performance.**



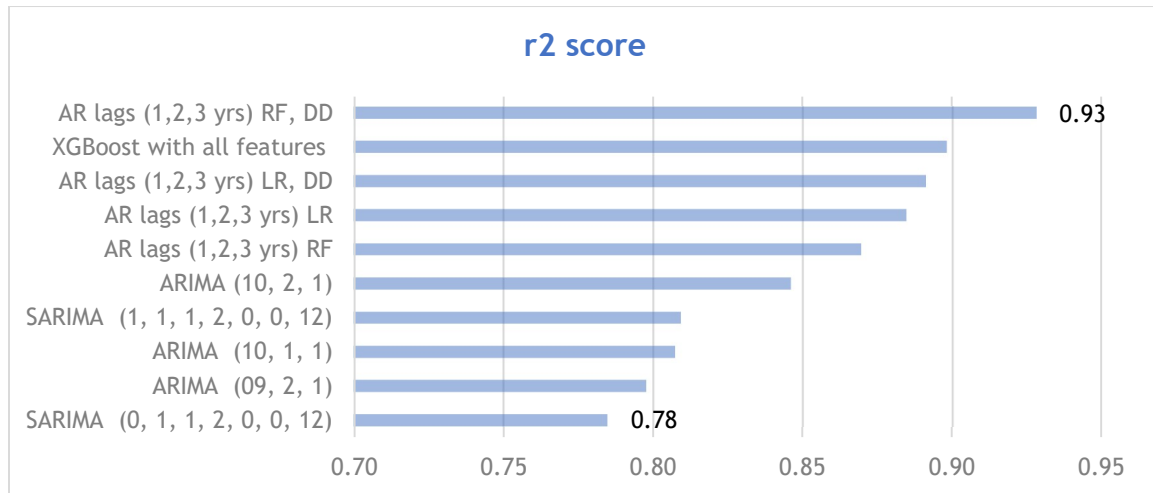
R-squared error of Coefficient of determination (R²) - measures the proportion of variance in the dependent variable (actual values) that is predictable from the independent variable (predicted values). It ranges from 0 to 1. Higher values indicate better prediction performance.

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

y_i = actual values

$\hat{y}_i$ (cap) = predicted values

$\bar{y}$ = mean of the actual values



Inference - So far Auto Regression with lags 1, 2, 3, years and m/c learning model Random Forest shows the best performance values.

12. Next steps (Section 4)

More exploration is required here.

- Explore SARIMA with grid search
- Explore SARIMAX with grid search and exogenous variables (additional features like season, month, population, temperature, Real Disposable Personal Income (RDPI) etc.)
- SARIMA, SARIMAX, XGBoost on decomposed data
- XGBoost with tuning the various parameters
- SARIMA, SARIMAX, XGBoost on decomposed data
- XGBoost with cross validation

New score with average of the best three scores, prediction,

13. Acronyms

Serial number	Acronym	Comments
1.	ANFIS	Adaptive Neuro Fuzzy Inference System
2.	ARIMA	Auto Regressive Integrated Moving Average
3.	ARMA	Auto Regressive Moving Average
4.	BPTT	Back Propagation Through Time
5.	CDD	Cooling Degree Days
6.	DE	Differential Evolutionary algorithm
7.	HDD	Heating Degree Days
8.	kWH	Kilo Watt-Hour
9.	LSTM	Long Short-Term Model
10.	MAM	Moving Average Model
11.	MAPE	Mean Absolute Percentage Error
12.	ML	Machine Learning
13.	MSE	Mean Squared Errors
14.	NN	Neural Network
15.	RDPI	Real Disposable Personnel Income
16.	RF	Random Forest
17.	RNN	Recurrent Neural Network
18.	SVR	Support Vector Regression
19.	VAR	Vector Auto Regression

14. References

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- Monthly electricity consumption and generation in USA for fifty years, [US Energy Information Administration](#)
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